Eye Tracking (Gaze) - Accuracy Methodology & Report

Model Evaluated: `src/eye_tracking.py` (MediaPipe Face Mesh + OpenCV Perspective-n-Point) **System Mechanism:** 3D Head Pose Yaw/Pitch tracking via `cv2.solvePnP` mapping to a 2D plane. **Test Dataset Status:** *Awaiting High-Resolution Gaze Dataset (e.g., MPIIGaze or BIWI Kinect)*

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1. Engine Core Difference (vs Emotion Model)

Unlike the emotion `.h5` model which accepts raw pixel maps into a Convolutional Neural Net, the Eye Tracker algorithm is **mathematical and geometric**. It processes the absolute coordinate matrices generated by `MediaPipe Face Mesh` (468 facial landmarks) mapped against a static camera depth projection. It specifically isolates:

- `Nose Tip: Landmark 1`
- `Chin: Landmark 152`
- `Eyes: Landmarks 33, 263`
- `Mouth Corners: Landmarks 61, 291`

As a result, **it cannot be evaluated on low-resolution cropped datasets (like FER-2013)**. MediaPipe requires high-resolution (preferably HD 720p or 1080p) uncropped facial images to accurately trace millimeters of pupil and chin shift.

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2. Evaluation Logic Extracted from your Code

The exported script `evaluate_eye_tracking.py` flawlessly maps to your exact logic matrix internally:

1. **Camera Matrix Construction:** Initializes a synthetic camera matrix mapping depth (`z-index * 3000`).
2. **Rotational Output (RQDecomp3x3):** Converts 3D rotation vectors into 2D Euler Angles (`Pitch` , `Yaw` , `Roll`).
3. **Threshold Classifications (The Labels):**
   - **`down`** : Pitch `> -10°`
   - **`up`** : Pitch `< 10°`
   - **`left`** : Yaw `< -10°`
   - **`right`** : Yaw `> 10°`
   - **`center`** : Pitch & Yaw between `-10°` and `10°`

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3. How to Calculate the Final Metrics

Because no appropriate dataset exists currently in the project, actual percentages have not been locked.

When you obtain a high-resolution gaze dataset (sorted into folders named `center` , `up` , `down` , `left` , `right`), run the newly created evaluator script:

```
python evaluate_eye_tracking.py --dataset path/to/gaze_data
```

This will automatically output the metrics that mimic the structure below.

Example Expected Output Matrix

Gaze Direction	Precision	Recall	F1-Score	Support

center (Looking at Camera)	-	-	-	-
up	-	-	-	-
down	-	-	-	-
left	-	-	-	-
right	-	-	-	-
Global Accuracy			**--.--%**	
MediaPipe Face Detection Rate			**--.--%**	

(By design, the classification report and confusion matrix structure will structurally mirror the Emotion Accuracy report, but will include a "Face Detection Rate", since `solvePnP` cannot return an angle if MediaPipe fails to detect a face algorithmically).